

Course Syllabus: Physics 225 – General Relativity 1

I. COURSE DESCRIPTION

Description: Manifolds, modern differential geometry and tensor analysis; basic principles of general relativity; Einstein field equations and their mathematical properties; exact solutions; linearized theory; variational principles and conservation laws; equations of motion; gravitational waves; experimental tests.

Prerequisite: Consent of instructor (preferably 221 and 231)

Credits: 3.0

II. CLASS AND INSTRUCTOR DETAILS

Instructor: Ian Vega <ivega@nip.upd.edu.ph>

Grader: Sean Fortuna <scfortuna@yahoo.com.ph>

Lecture hours: WF, 4:00-5:30 PM.

III. TEXTBOOK AND USEFUL REFERENCES

We shall not have a main textbook for the class. I used *Schutz*, supplemented by the first edition of *d’Inverno*, when I first started learning GR. I think they are both very good texts. However, I have my own way of presenting GR that will not always follow the structure or presentation of any single book. My approach is closest in style to the treatments of *Liang+Zhou* and *Wald*.

There are numerous sources you can refer to. I highly recommend that you look at different texts to help understand my lectures better. Here are some good books I’ve used on numerous occasions.

1. R d’Inverno, J Vickers, *Introducing Einstein’s Relativity: A Deeper Understanding*, 2022
2. C Liang and B Zhou, *Differential Geometry and General Relativity*, 2023
3. R Wald, *General Relativity*, 1984
4. V Ferrari, L Gualtieri, P Pani, *General Relativity and its Applications*, 2021
5. T Padmanabhan, *Gravitation: Foundations and Frontiers*, 2010
6. A Zee, *Einstein Gravity in a Nutshell*, 2013.
7. S Carroll, *Spacetime and Geometry: An Introduction to General Relativity*, 2003
8. Choquet-Bruhat, *Introduction to General Relativity, Black Holes, and Cosmology*, 2015
9. J Hartle, *Gravity: An Introduction to Einstein’s General Relativity*, 2003. (undergrad)
10. B Schutz, *A First Course in General Relativity* 2e, 2009. (undergrad)
11. MP Hobson, GP Efstathiou, and AN Lasenby, *General Relativity: An Introduction for Physicists*, 2006. (strong for cosmology)

For differential geometry:

1. B O’Neill, *Semi-Riemannian Geometry*, 1983
2. S Lovett, *Differential Geometry of Manifolds* 2nd Ed, 2020

IV. COURSE OUTLINE

1. Review of Newtonian physics and special relativity
2. Aspects of differential geometry: manifolds, tensors, differentiation, connections, curvature
3. Einstein field equation
4. Matter models, stress-energy tensors: perfect fluids, point particles, electromagnetism
5. Weak-field solutions: Newtonian gravity, gravitational waves
6. Spherical symmetry: relativistic stars and the Schwarzschild solution
7. Basic cosmology

V. COURSE REQUIREMENTS

Problem Sets: Problem sets will be assigned throughout the semester, ideally one per week. These will be submitted electronically using a Google Form that I will be sending out close to the deadline. From each problem set a random selection of items will be graded. It is your responsibility to ensure that your work is neat and legible. If we cannot understand your work then you will not get any credit. These problem sets will make up the 30% of your final grade.

Written Exams: There will be two written exams in this course. The first will be on 24 March 2025 (Monday), 1-5 pm. The second will be on 23 May 2025 (Friday), 1-5pm. Each exam will constitute 30% of your final grade. These are in-class, open-notes exams. You will have to designate one notebook as your 225 notebook at the start of the semester. Only this notebook may be used to help you during the exams.

Oral Exam: This will be scheduled during or shortly after Finals Week. This will make up the remaining 10% of your final grade.

VI. SOME POLICIES

Collaboration on problem sets: Collaboration is highly encouraged. Material is often learned better when it is made part of active discussion. However, you are to write up your problem set solutions entirely on your own and, as is customary in scientific practice, you must clearly acknowledge all your collaborators in your submitted work.

Late problem sets: Problem sets may be submitted up to one week late. However, any late submissions can only go as high as 70%. (I will multiply a factor of 0.7 to the raw score of any late submission.)

Make-up exams: I will give make-up exams to *excused* absences. This will generally require a legitimate medical certificate. Make-up exams will be given sometime during the Finals Week (May 24-30).

Policy on INC grade: Should you not be able to complete the course in any way (e.g. you are unable to take make-up exams for excused absences during the Finals Week), then you will get an INC. You will have up to one year to take the exams and complete the course, but the highest grade you can get upon completion cannot exceed the lowest grade among the students who completed the class in the regular semester.

VII. GRADING

x	Point grade
$100\% \geq x \geq 90\%$	1.00
$90\% > x \geq 85\%$	1.25
$85\% > x \geq 80\%$	1.50
$80\% > x \geq 75\%$	1.75
$75\% > x \geq 70\%$	2.00
$70\% > x \geq 65\%$	2.25
$65\% > x \geq 60\%$	2.50
$60\% > x \geq 55\%$	2.75
$55\% > x \geq 50\%$	3.00
$50\% > x \geq 45\%$	4.00
$45\% > x$	5.00